Using this interpretation we can paraphrase weak duality as follows: The optimal cost to the firm in the second scenario (in which constraint violations can be bought and sold) is less than or equal to the cost in the original situation (which has constraints that cannot be violated), even with the most unfavorable prices. This is obvious: If $x^\star$ is optimal in the first scenario, then the operating cost of $x^\star$ in the second scenario will be lower than $f_0(x^\star)$, since some income can be derived from the constraints that are not tight. The optimal duality gap is then the minimum possible advantage to the enterprise of being allowed to pay for constraint violations (and receive payments for nontight constraints).

Now suppose strong duality holds, and the dual optimum is attained. We can interpret a dual optimal $\lambda^\star$ as a set of prices for which there is no advantage to the firm in being allowed to pay for constraint violations (or receive payments for nontight constraints). For this reason a dual optimal $\lambda^\star$ is sometimes called a set of *shadow prices* for the original problem.

5.5 Optimality conditions

We remind the reader that we do not assume the problem (5.1) is convex, unless explicitly stated.

5.5.1 Certificate of suboptimality and stopping criteria

If we can find a dual feasible (λ, ν), we establish a lower bound on the optimal value of the primal problem: $p^\star \geq g(\lambda, \nu)$. Thus a dual feasible point (λ, ν) provides a *proof* or *certificate* that $p^\star \geq g(\lambda, \nu)$. Strong duality means there exist arbitrarily good certificates.

Dual feasible points allow us to bound how suboptimal a given feasible point is, without knowing the exact value of $p^\star$. Indeed, if x is primal feasible and (λ, ν) is dual feasible, then

$$f_0(x) - p^\star \leq f_0(x) - g(\lambda, \nu).$$

In particular, this establishes that x is ϵ-suboptimal, with $\epsilon = f_0(x) - g(\lambda, \nu)$. (It also establishes that (λ, ν) is ϵ-suboptimal for the dual problem.)

We refer to the gap between primal and dual objectives,

$$f_0(x) - g(\lambda, \nu),$$

as the *duality gap* associated with the primal feasible point x and dual feasible point (λ, ν). A primal dual feasible pair x, (λ, ν) localizes the optimal value of the primal (and dual) problems to an interval:

$$p^\star \in [g(\lambda, \nu), f_0(x)], \qquad d^\star \in [g(\lambda, \nu), f_0(x)],$$

the width of which is the duality gap.

If the duality gap of the primal dual feasible pair x, (λ, ν) is zero, *i.e.*, $f_0(x) = g(\lambda, \nu)$, then x is primal optimal and (λ, ν) is dual optimal. We can think of (λ, ν)

as a certificate that proves x is optimal (and, similarly, we can think of x as a certificate that proves (λ, ν) is dual optimal).

These observations can be used in optimization algorithms to provide nonheuristic stopping criteria. Suppose an algorithm produces a sequence of primal feasible $x^{(k)}$ and dual feasible $(\lambda^{(k)}, \nu^{(k)})$, for $k = 1, 2, \ldots$, and $\epsilon_{\text{abs}} > 0$ is a given required absolute accuracy. Then the stopping criterion (*i.e.*, the condition for terminating the algorithm)

$$f_0(x^{(k)}) - g(\lambda^{(k)}, \nu^{(k)}) \leq \epsilon_{\text{abs}}$$

guarantees that when the algorithm terminates, $x^{(k)}$ is ϵ_{abs}-suboptimal. Indeed, $(\lambda^{(k)}, \nu^{(k)})$ is a certificate that proves it. (Of course strong duality must hold if this method is to work for arbitrarily small tolerances ϵ_{abs}.)

A similar condition can be used to guarantee a given relative accuracy $\epsilon_{\text{rel}} > 0$. If

$$g(\lambda^{(k)}, \nu^{(k)}) > 0, \qquad \frac{f_0(x^{(k)}) - g(\lambda^{(k)}, \nu^{(k)})}{g(\lambda^{(k)}, \nu^{(k)})} \leq \epsilon_{\text{rel}}$$

holds, or

$$f_0(x^{(k)}) < 0, \qquad \frac{f_0(x^{(k)}) - g(\lambda^{(k)}, \nu^{(k)})}{-f_0(x^{(k)})} \leq \epsilon_{\text{rel}}$$

holds, then $p^\star \neq 0$ and the relative error

$$\frac{f_0(x^{(k)}) - p^\star}{|p^\star|}$$

is guaranteed to be less than or equal to ϵ_{rel}.

5.5.2 Complementary slackness

Suppose that the primal and dual optimal values are attained and equal (so, in particular, strong duality holds). Let $x^\star$ be a primal optimal and $(\lambda^\star, \nu^\star)$ be a dual optimal point. This means that

$$\begin{aligned} f_0(x^\star) &= g(\lambda^\star, \nu^\star) \\ &= \inf_x \left(f_0(x) + \sum_{i=1}^m \lambda_i^\star f_i(x) + \sum_{i=1}^p \nu_i^\star h_i(x) \right) \\ &\leq f_0(x^\star) + \sum_{i=1}^m \lambda_i^\star f_i(x^\star) + \sum_{i=1}^p \nu_i^\star h_i(x^\star) \\ &\leq f_0(x^\star). \end{aligned}$$

The first line states that the optimal duality gap is zero, and the second line is the definition of the dual function. The third line follows since the infimum of the Lagrangian over x is less than or equal to its value at $x = x^\star$. The last inequality follows from $\lambda_i^\star \geq 0$, $f_i(x^\star) \leq 0$, $i = 1, \ldots, m$, and $h_i(x^\star) = 0$, $i = 1, \ldots, p$. We conclude that the two inequalities in this chain hold with equality.

We can draw several interesting conclusions from this. For example, since the inequality in the third line is an equality, we conclude that $x^\star$ minimizes $L(x, \lambda^\star, \nu^\star)$ over x. (The Lagrangian $L(x, \lambda^\star, \nu^\star)$ can have other minimizers; $x^\star$ is simply *a* minimizer.)

Another important conclusion is that

$$\sum_{i=1}^{m} \lambda_i^\star f_i(x^\star) = 0.$$

Since each term in this sum is nonpositive, we conclude that

$$\lambda_i^\star f_i(x^\star) = 0, \quad i = 1, \ldots, m. \tag{5.48}$$

This condition is known as *complementary slackness*; it holds for any primal optimal $x^\star$ and any dual optimal $(\lambda^\star, \nu^\star)$ (when strong duality holds). We can express the complementary slackness condition as

$$\lambda_i^\star > 0 \implies f_i(x^\star) = 0,$$

or, equivalently,

$$f_i(x^\star) < 0 \implies \lambda_i^\star = 0.$$

Roughly speaking, this means the ith optimal Lagrange multiplier is zero unless the ith constraint is active at the optimum.

5.5.3 KKT optimality conditions

We now assume that the functions $f_0, \ldots, f_m, h_1, \ldots, h_p$ are differentiable (and therefore have open domains), but we make no assumptions yet about convexity.

KKT conditions for nonconvex problems

As above, let $x^\star$ and $(\lambda^\star, \nu^\star)$ be any primal and dual optimal points with zero duality gap. Since $x^\star$ minimizes $L(x, \lambda^\star, \nu^\star)$ over x, it follows that its gradient must vanish at $x^\star$, *i.e.*,

$$\nabla f_0(x^\star) + \sum_{i=1}^{m} \lambda_i^\star \nabla f_i(x^\star) + \sum_{i=1}^{p} \nu_i^\star \nabla h_i(x^\star) = 0.$$

Thus we have

$$\begin{array}{rcll} f_i(x^\star) & \leq & 0, & i = 1, \ldots, m \\ h_i(x^\star) & = & 0, & i = 1, \ldots, p \\ \lambda_i^\star & \geq & 0, & i = 1, \ldots, m \\ \lambda_i^\star f_i(x^\star) & = & 0, & i = 1, \ldots, m \\ \nabla f_0(x^\star) + \sum_{i=1}^{m} \lambda_i^\star \nabla f_i(x^\star) + \sum_{i=1}^{p} \nu_i^\star \nabla h_i(x^\star) & = & 0, & \end{array} \tag{5.49}$$

which are called the *Karush-Kuhn-Tucker* (KKT) conditions.

To summarize, for *any* optimization problem with differentiable objective and constraint functions for which strong duality obtains, any pair of primal and dual optimal points must satisfy the KKT conditions (5.49).

KKT conditions for convex problems

When the primal problem is convex, the KKT conditions are also sufficient for the points to be primal and dual optimal. In other words, if f_i are convex and h_i are affine, and $\tilde{x}$, $\tilde{\lambda}$, $\tilde{\nu}$ are any points that satisfy the KKT conditions

$$\begin{array}{rcll} f_i(\tilde{x}) & \leq & 0, & i = 1, \ldots, m \\ h_i(\tilde{x}) & = & 0, & i = 1, \ldots, p \\ \tilde{\lambda}_i & \geq & 0, & i = 1, \ldots, m \\ \tilde{\lambda}_i f_i(\tilde{x}) & = & 0, & i = 1, \ldots, m \\ \nabla f_0(\tilde{x}) + \sum_{i=1}^m \tilde{\lambda}_i \nabla f_i(\tilde{x}) + \sum_{i=1}^p \tilde{\nu}_i \nabla h_i(\tilde{x}) & = & 0, & \end{array}$$

then $\tilde{x}$ and $(\tilde{\lambda}, \tilde{\nu})$ are primal and dual optimal, with zero duality gap.

To see this, note that the first two conditions state that $\tilde{x}$ is primal feasible. Since $\tilde{\lambda}_i \geq 0$, $L(x, \tilde{\lambda}, \tilde{\nu})$ is convex in x; the last KKT condition states that its gradient with respect to x vanishes at $x = \tilde{x}$, so it follows that $\tilde{x}$ minimizes $L(x, \tilde{\lambda}, \tilde{\nu})$ over x. From this we conclude that

$$\begin{aligned} g(\tilde{\lambda}, \tilde{\nu}) &= L(\tilde{x}, \tilde{\lambda}, \tilde{\nu}) \\ &= f_0(\tilde{x}) + \sum_{i=1}^m \tilde{\lambda}_i f_i(\tilde{x}) + \sum_{i=1}^p \tilde{\nu}_i h_i(\tilde{x}) \\ &= f_0(\tilde{x}), \end{aligned}$$

where in the last line we use $h_i(\tilde{x}) = 0$ and $\tilde{\lambda}_i f_i(\tilde{x}) = 0$. This shows that $\tilde{x}$ and $(\tilde{\lambda}, \tilde{\nu})$ have zero duality gap, and therefore are primal and dual optimal. In summary, for any *convex* optimization problem with differentiable objective and constraint functions, any points that satisfy the KKT conditions are primal and dual optimal, and have zero duality gap.

If a convex optimization problem with differentiable objective and constraint functions satisfies Slater's condition, then the KKT conditions provide necessary and sufficient conditions for optimality: Slater's condition implies that the optimal duality gap is zero and the dual optimum is attained, so x is optimal if and only if there are (λ, ν) that, together with x, satisfy the KKT conditions.

The KKT conditions play an important role in optimization. In a few special cases it is possible to solve the KKT conditions (and therefore, the optimization problem) analytically. More generally, many algorithms for convex optimization are conceived as, or can be interpreted as, methods for solving the KKT conditions.

Example 5.1 *Equality constrained convex quadratic minimization.* We consider the problem

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P x + q^T x + r \\ \text{subject to} & Ax = b, \end{array} \tag{5.50}$$

where $P \in \mathbf{S}_+^n$. The KKT conditions for this problem are

$$Ax^\star = b, \qquad Px^\star + q + A^T \nu^\star = 0,$$

which we can write as

$$\begin{bmatrix} P & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} x^\star \\ \nu^\star \end{bmatrix} = \begin{bmatrix} -q \\ b \end{bmatrix}.$$

Solving this set of $m + n$ equations in the $m + n$ variables $x^\star$, $\nu^\star$ gives the optimal primal and dual variables for (5.50).

Example 5.2 *Water-filling.* We consider the convex optimization problem

$$\begin{array}{ll} \text{minimize} & -\sum_{i=1}^n \log(\alpha_i + x_i) \\ \text{subject to} & x \succeq 0, \quad \mathbf{1}^T x = 1, \end{array}$$

where $\alpha_i > 0$. This problem arises in information theory, in allocating power to a set of n communication channels. The variable x_i represents the transmitter power allocated to the ith channel, and $\log(\alpha_i + x_i)$ gives the capacity or communication rate of the channel, so the problem is to allocate a total power of one to the channels, in order to maximize the total communication rate.

Introducing Lagrange multipliers $\lambda^\star \in \mathbf{R}^n$ for the inequality constraints $x^\star \succeq 0$, and a multiplier $\nu^\star \in \mathbf{R}$ for the equality constraint $\mathbf{1}^T x = 1$, we obtain the KKT conditions

$$x^\star \succeq 0, \qquad \mathbf{1}^T x^\star = 1, \qquad \lambda^\star \succeq 0, \qquad \lambda_i^\star x_i^\star = 0, \quad i = 1, \ldots, n,$$

$$-1/(\alpha_i + x_i^\star) - \lambda_i^\star + \nu^\star = 0, \quad i = 1, \ldots, n.$$

We can directly solve these equations to find $x^\star$, $\lambda^\star$, and $\nu^\star$. We start by noting that $\lambda^\star$ acts as a slack variable in the last equation, so it can be eliminated, leaving

$$x^\star \succeq 0, \qquad \mathbf{1}^T x^\star = 1, \qquad x_i^\star \left(\nu^\star - 1/(\alpha_i + x_i^\star)\right) = 0, \quad i = 1, \ldots, n,$$

$$\nu^\star \geq 1/(\alpha_i + x_i^\star), \quad i = 1, \ldots, n.$$

If $\nu^\star < 1/\alpha_i$, this last condition can only hold if $x_i^\star > 0$, which by the third condition implies that $\nu^\star = 1/(\alpha_i + x_i^\star)$. Solving for $x_i^\star$, we conclude that $x_i^\star = 1/\nu^\star - \alpha_i$ if $\nu^\star < 1/\alpha_i$. If $\nu^\star \geq 1/\alpha_i$, then $x_i^\star > 0$ is impossible, because it would imply $\nu^\star \geq 1/\alpha_i > 1/(\alpha_i + x_i^\star)$, which violates the complementary slackness condition. Therefore, $x_i^\star = 0$ if $\nu^\star \geq 1/\alpha_i$. Thus we have

$$x_i^\star = \begin{cases} 1/\nu^\star - \alpha_i & \nu^\star < 1/\alpha_i \\ 0 & \nu^\star \geq 1/\alpha_i, \end{cases}$$

or, put more simply, $x_i^\star = \max\{0, 1/\nu^\star - \alpha_i\}$. Substituting this expression for $x_i^\star$ into the condition $\mathbf{1}^T x^\star = 1$ we obtain

$$\sum_{i=1}^n \max\{0, 1/\nu^\star - \alpha_i\} = 1.$$

The lefthand side is a piecewise-linear increasing function of $1/\nu^\star$, with breakpoints at α_i, so the equation has a unique solution which is readily determined.

This solution method is called *water-filling* for the following reason. We think of α_i as the ground level above patch i, and then flood the region with water to a depth $1/\nu$, as illustrated in figure 5.7. The total amount of water used is then $\sum_{i=1}^n \max\{0, 1/\nu^\star - \alpha_i\}$. We then increase the flood level until we have used a total amount of water equal to one. The depth of water above patch i is then the optimal value $x_i^\star$.

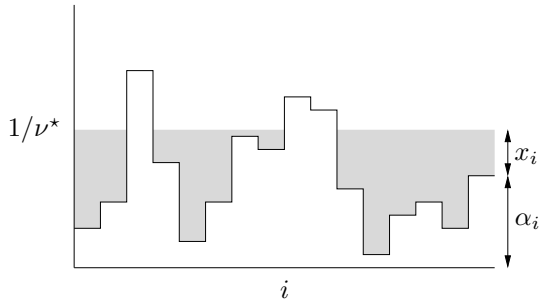


Figure 5.7 Illustration of water-filling algorithm. The height of each patch is given by α_i. The region is flooded to a level $1/\nu^\star$ which uses a total quantity of water equal to one. The height of the water (shown shaded) above each patch is the optimal value of $x_i^\star$.

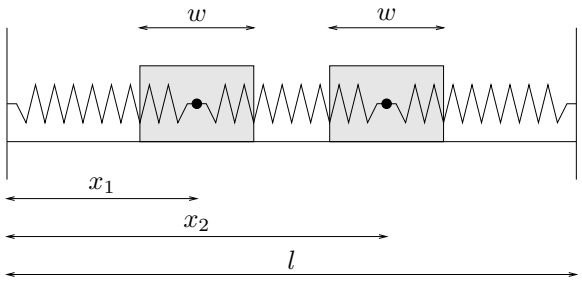


Figure 5.8 Two blocks connected by springs to each other, and the left and right walls. The blocks have width $w > 0$, and cannot penetrate each other or the walls.

5.5.4 Mechanics interpretation of KKT conditions

The KKT conditions can be given a nice interpretation in mechanics (which indeed, was one of Lagrange's primary motivations). We illustrate the idea with a simple example. The system shown in figure 5.8 consists of two blocks attached to each other, and to walls at the left and right, by three springs. The position of the blocks are given by $x \in \mathbf{R}^2$, where x_1 is the displacement of the (middle of the) left block, and x_2 is the displacement of the right block. The left wall is at position 0, and the right wall is at position l.

The potential energy in the springs, as a function of the block positions, is given by

$$f_0(x_1, x_2) = \frac{1}{2}k_1x_1^2 + \frac{1}{2}k_2(x_2 - x_1)^2 + \frac{1}{2}k_3(l - x_2)^2,$$

where $k_i > 0$ are the stiffness constants of the three springs. The equilibrium position $x^\star$ is the position that minimizes the potential energy subject to the inequalities

$$w/2 - x_1 \leq 0, \qquad w + x_1 - x_2 \leq 0, \qquad w/2 - l + x_2 \leq 0. \tag{5.51}$$

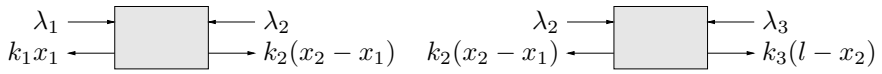


Figure 5.9 Force analysis of the block-spring system. The total force on each block, due to the springs and also to contact forces, must be zero. The Lagrange multipliers, shown on top, are the contact forces between the walls and blocks. The spring forces are shown at bottom.

These constraints are called *kinematic constraints*, and express the fact that the blocks have width $w > 0$, and cannot penetrate each other or the walls. The equilibrium position is therefore given by the solution of the optimization problem

$$\begin{array}{ll} \text{minimize} & (1/2)\left(k_1x_1^2 + k_2(x_2 - x_1)^2 + k_3(l - x_2)^2\right) \\ \text{subject to} & w/2 - x_1 \leq 0 \\ & w + x_1 - x_2 \leq 0 \\ & w/2 - l + x_2 \leq 0, \end{array} \tag{5.52}$$

which is a QP.

With λ_1, λ_2, λ_3 as Lagrange multipliers, the KKT conditions for this problem consist of the kinematic constraints (5.51), the nonnegativity constraints $\lambda_i \geq 0$, the complementary slackness conditions

$$\lambda_1(w/2 - x_1) = 0, \qquad \lambda_2(w - x_2 + x_1) = 0, \qquad \lambda_3(w/2 - l + x_2) = 0, \tag{5.53}$$

and the zero gradient condition

$$\begin{bmatrix} k_1x_1 - k_2(x_2 - x_1) \\ k_2(x_2 - x_1) - k_3(l - x_2) \end{bmatrix} + \lambda_1 \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0. \tag{5.54}$$

The equation (5.54) can be interpreted as the force balance equations for the two blocks, provided we interpret the Lagrange multipliers as *contact forces* that act between the walls and blocks, as illustrated in figure 5.9. The first equation states that the sum of the forces on the first block is zero: The term $-k_1x_1$ is the force exerted on the left block by the left spring, the term $k_2(x_2 - x_1)$ is the force exerted by the middle spring, λ_1 is the force exerted by the left wall, and $-\lambda_2$ is the force exerted by the right block. The contact forces must point away from the contact surface (as expressed by the constraints $\lambda_1 \geq 0$ and $-\lambda_2 \leq 0$), and are nonzero only when there is contact (as expressed by the first two complementary slackness conditions (5.53)). In a similar way, the second equation in (5.54) is the force balance for the second block, and the last condition in (5.53) states that λ_3 is zero unless the right block touches the wall.

In this example, the potential energy and kinematic constraint functions are convex, and (the refined form of) Slater's constraint qualification holds provided $2w \leq l$, *i.e.*, there is enough room between the walls to fit the two blocks, so we can conclude that the energy formulation of the equilibrium given by (5.52), gives the same result as the force balance formulation, given by the KKT conditions.

5.5.5 Solving the primal problem via the dual

We mentioned at the beginning of §5.5.3 that if strong duality holds and a dual optimal solution $(\lambda^\star, \nu^\star)$ exists, then any primal optimal point is also a minimizer of $L(x, \lambda^\star, \nu^\star)$. This fact sometimes allows us to compute a primal optimal solution from a dual optimal solution.

More precisely, suppose we have strong duality and an optimal $(\lambda^\star, \nu^\star)$ is known. Suppose that the minimizer of $L(x, \lambda^\star, \nu^\star)$, *i.e.*, the solution of

$$\text{minimize} \quad f_0(x) + \sum_{i=1}^m \lambda_i^\star f_i(x) + \sum_{i=1}^p \nu_i^\star h_i(x), \tag{5.55}$$

is unique. (For a convex problem this occurs, for example, if $L(x, \lambda^\star, \nu^\star)$ is a strictly convex function of x.) Then if the solution of (5.55) is primal feasible, it must be primal optimal; if it is not primal feasible, then no primal optimal point can exist, *i.e.*, we can conclude that the primal optimum is not attained. This observation is interesting when the dual problem is easier to solve than the primal problem, for example, because it can be solved analytically, or has some special structure that can be exploited.

Example 5.3 *Entropy maximization.* We consider the entropy maximization problem

$$\begin{array}{ll} \text{minimize} & f_0(x) = \sum_{i=1}^n x_i \log x_i \\ \text{subject to} & Ax \preceq b \\ & \mathbf{1}^T x = 1 \end{array}$$

with domain $\mathbf{R}^n_{++}$, and its dual problem

$$\begin{array}{ll} \text{maximize} & -b^T\lambda - \nu - e^{-\nu-1} \sum_{i=1}^n e^{-a_i^T\lambda} \\ \text{subject to} & \lambda \succeq 0 \end{array}$$

where a_i are the columns of A (see pages 222 and 228). We assume that the weak form of Slater's condition holds, *i.e.*, there exists an $x \succ 0$ with $Ax \preceq b$ and $\mathbf{1}^T x = 1$, so strong duality holds and an optimal solution $(\lambda^\star, \nu^\star)$ exists.

Suppose we have solved the dual problem. The Lagrangian at $(\lambda^\star, \nu^\star)$ is

$$L(x, \lambda^\star, \nu^\star) = \sum_{i=1}^n x_i \log x_i + \lambda^{\star T}(Ax - b) + \nu^\star(\mathbf{1}^T x - 1)$$

which is strictly convex on $\mathcal{D}$ and bounded below, so it has a unique solution $x^\star$, given by

$$x_i^\star = 1/\exp(a_i^T\lambda^\star + \nu^\star + 1), \quad i = 1, \ldots, n.$$

If $x^\star$ is primal feasible, it must be the optimal solution of the primal problem (5.13). If $x^\star$ is not primal feasible, then we can conclude that the primal optimum is not attained.

Example 5.4 *Minimizing a separable function subject to an equality constraint.* We consider the problem

$$\begin{array}{ll} \text{minimize} & f_0(x) = \sum_{i=1}^n f_i(x_i) \\ \text{subject to} & a^T x = b, \end{array}$$